Digital Inclusion and Circular Economy Behavior in Indian States: A Digital Circularity Index and Explainable XGBoost Analysis

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Abstract: India's digital build-out and circular-economy agenda are often discussed in separate policy spaces: one in terms of broadband, mobile access and platform payments, the other in terms of waste, resource efficiency and environmental performance. At the state level, however, both transitions depend on a shared capacity to collect data, coordinate services and make resource flows visible. This paper asks whether states with stronger digital access also show stronger circular-economy behavior. Using seven official and policy-relevant data sources, we compile a harmonised dataset for 16 Indian states and construct three indicators: the Digital Access Score (DAS), the Circular Economy Behavior Score (CEBS) and the Digital Circularity Index (DCI). The analysis combines descriptive comparison, ordinary least squares regression, rank-based and permutation-based checks, mediation and threshold testing, leak-free nested cross-validation, SHAP-based explainability and K-Means clustering. The association is consistent but is interpreted cautiously. DAS and CEBS are positively related ($r = 0.781$, $p < 0.001$), and the result remains visible under alternative inference checks. Tele-density carries more explanatory weight than UPI adoption, which does not mediate the main relationship. This suggests that digital inclusion cannot be reduced to payment activity alone. Nested leave-one-out cross-validation gives XGBoost the strongest leak-free result ($R^2 = 0.511$), while clustering separates digitally stronger states from states where both digital and circularity indicators remain weaker. We read these findings as evidence of linked development capacity, not as proof of causality. For Indian policy, the practical implication is that connectivity, reporting systems and circular-economy administration should be planned together, especially in states where foundational capacity is weakest.

Keywords: circular economy; digital circularity index (DCI); digital inclusion; explainable machine learning; Indian states; XGBoost

1 Introduction

India's digital transition is now visible in everyday administration and commerce. Monthly UPI transactions exceed 14 billion (NPCI, 2025), national tele-density remains above 85% (TRAI, 2024), and the Digital India programme has pushed broadband and mobile services into a wide range of regional settings. A second transition is unfolding at the same time. India is under pressure to manage e-waste, urban waste and resource-use inefficiency, while national policy institutions increasingly treat circularity as part of long-term growth strategy (CPCB, 2023; NITI Aayog, 2021).

The overlap matters because both agendas rely on more than technology adoption. Digital systems help only when they improve reporting, reach and coordination; circular-economy programmes also need reliable information about collection, processing and resource movement. Yet research on digital inclusion in India still concentrates mainly on access, payments and growth (Ministry of Electronics & IT, 2023), while circular-economy research tends to focus on waste systems, environmental governance and industrial transition (Geissdoerfer et al., 2017). State-level evidence connecting the two remains thin. This paper builds a Digital Circularity Index from seven public data sources and tests whether digitally stronger states also perform better on selected circular-economy indicators.

The study evaluates three hypotheses. H1 states that states with higher Digital Access Scores will also have higher CE Behavior Scores. H2 asks whether UPI adoption acts as the channel between digital access and CE behavior. H3 tests whether CE behavior changes more sharply after a tele-density threshold.

2 Literature Review

Digital inclusion in India has advanced rapidly, but not evenly. Tele-density and digital payments have expanded across the country, yet large gaps remain between higher-capacity states and states with weaker infrastructure or lower income levels (TRAI, 2024; NPCI, 2025). Recent subnational work makes the same point: Indian states are moving through digitalisation at different speeds, with some states combining broad access and balanced use while others still face foundational constraints (Mishra et al., 2025). Much of the policy literature treats these developments as routes to financial inclusion, efficiency and service access (Ministry of Electronics & IT, 2023). ICT-for-development scholarship is broader. It argues that digital systems also reshape governance, information flows and institutional capacity, which means their effects may extend into environmental management and sustainability outcomes (Avgerou, 2010; Heeks, 2017).

Circular-economy research points to a similar implementation problem. The concept has moved from global advocacy into Indian policy planning, especially through NITI Aayog's sector-based framework (Ellen MacArthur Foundation, 2013; NITI Aayog, 2021). Even so, formal e-waste processing remains limited, state performance varies, and SDG-linked outcomes still differ substantially (CPCB, 2023; NITI Aayog, 2024). Recent work on digital technologies and CE also places emphasis on information functions: generating data, tracking product origins, supporting resource management and making material loops easier to manage (Schöggl et al., 2023). Taken together, these strands suggest a plausible connection. States with stronger digital access may be better positioned to coordinate reporting, traceability and circular-economy behavior. This paper tests that possibility directly.

3 Methodology

3.1 Research design and data

The study uses a cross-sectional design based on 16 Indian states with complete observations across all selected inputs. Because the sources do not cover every state in the same way, the sample is deliberately conservative: we keep only states that are comparable across every source and prioritise traceability over sample size. The final dataset combines nine raw and six derived variables, and Table 1 lists the core source variables.

Table 1: Data Sources and Core Variables

Source	Variable	Year
TRAI	Tele-density (%)	2024
RBI	Per capita NSDP (INR)	2023–24
NITI Aayog	SDG Index Score	2023–24
NPCI/Dataful Insights	UPI transactions per capita	2024–25
CPCB	E-waste processed (tonnes)	2021–22
Census of India	Literacy rate (%)	2011
Census of India	Urban population (%)	2011

Source: Compiled by the authors from official public datasets.

3.2 Composite index construction

To keep the index construction transparent, each raw variable is first rescaled to a 0–100 range using min–max normalisation:

$$\hat{x}_i = \frac{x_i - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}} \times 100 \tag{1}$$

The Digital Access Score is defined as:

$$DAS = 0.40\hat{T}_d + 0.30\hat{U}_{pc} + 0.30\hat{N}_{pc} \tag{2}$$

where $\hat{T}_d$ is tele-density, $\hat{U}_{pc}$ is UPI transactions per capita and $\hat{N}_{pc}$ is per capita NSDP after normalisation.
The Circular Economy Behavior Score is defined as:

$$CEBS = 0.35\hat{E}_{pc} + 0.40S_{sdg} + 0.25\hat{N}_{pc} \tag{3}$$

where $\hat{E}_{pc}$ is normalised e-waste processed per capita and S_{sdg} is the SDG Index score. The Digital Circularity Index then combines both dimensions multiplicatively:

$$DCI = \frac{DAS \times CEBS}{100} \tag{4}$$

3.3 Analytical strategy

H1 is tested using a bivariate OLS model of CEBS on DAS, followed by HC3 robust errors, permutation tests, bootstrap intervals and rank-based correlations. H2 is assessed using the Baron and Kenny mediation framework with a Sobel test (Baron and Kenny, 1986). H3 is examined through piecewise regression with grid-searched breakpoint selection. Predictive stability is checked first with leave-one-out cross-validation and then with a stricter nested leave-one-out design. The machine-learning step is used as a stress test of the pattern rather than as a claim of operational forecasting accuracy. XGBoost is interpreted using SHAP values (Lundberg and Lee, 2017), and K-Means clustering is used to identify state typologies.

4 Results and Discussion

4.1 Descriptive patterns

The first pattern is simple: DCI is not spread evenly across the sample. Haryana records the highest DCI (72.97), Karnataka posts the highest DAS (94.40), and Bihar ranks lowest across all three composite indicators. Figure 1 shows the state-level spread.

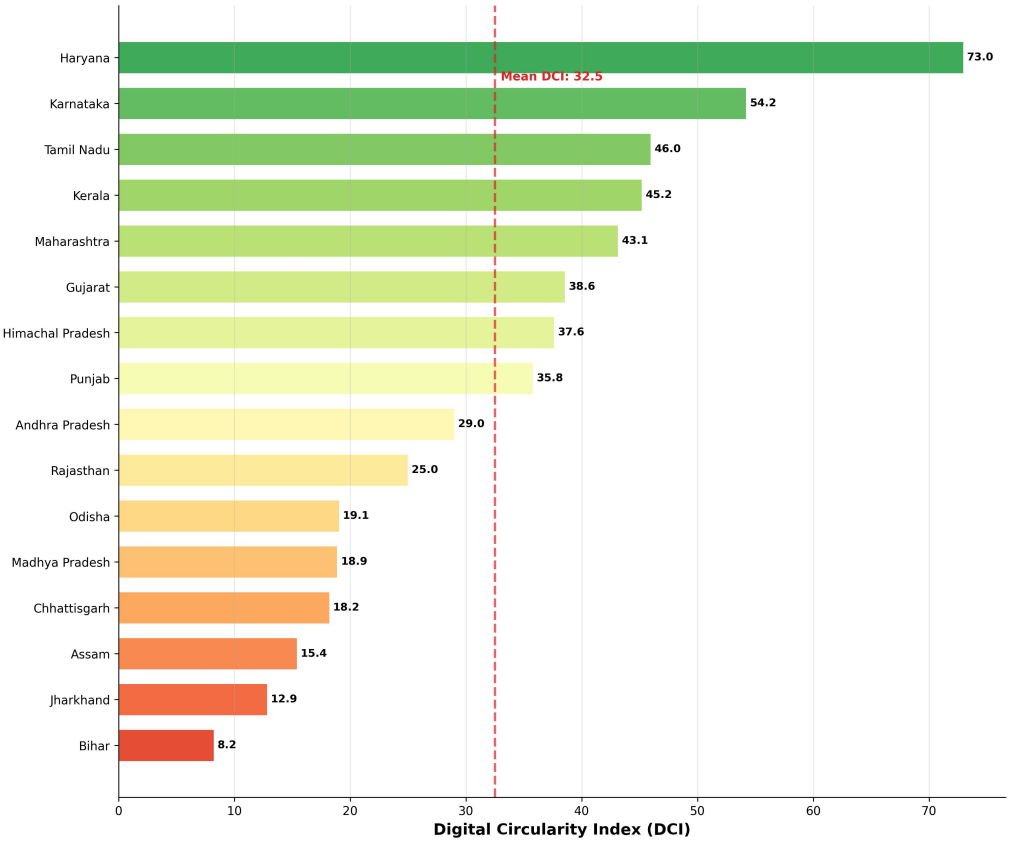


Figure 1: Digital Circularity Index ranking across 16 Indian states.

Source: Authors' analysis based on the compiled state-level dataset.

4.2 Regression, robustness and mechanisms

The regression confirms that the visual pattern is not only descriptive. A one-point increase in DAS is associated with a 0.551-point increase in CEBS, and the model explains 61% of the observed variation. Figure 2 shows the positive association, while Table 2 summarises the main findings. The relationship remains significant under HC3 robust inference, permutation testing and rank-based correlation, which reduces concern that the result is only an artefact of distributional assumptions.

The UPI result is useful precisely because it is negative. Once DAS is already in the model, UPI adoption contributes little explanatory value and the indirect path is not significant. Digital access in this sample therefore appears to matter less as payment activity and more as a wider infrastructure and capacity condition. By contrast, the threshold analysis offers qualified support for H3. CE performance improves up to roughly 119% tele-density, but the post-threshold slope is estimated from only a few states, so the breakpoint should be interpreted cautiously.

4.3 Predictive validation and explainability

The predictive analysis asks a modest question: does the same signal remain when each state is held out in turn? In standard leave-one-out cross-validation, ElasticNet performs best among the digital-only models ($R^2 = 0.364$), while Lasso performs best in the full-feature comparison ($R^2 = 0.568$). Under the stricter nested leave-one-out setup, XGBoost achieves the strongest leak-free result ($R^2 = 0.511$), narrowly outperforming Random Forest. SHAP analysis then points to tele-density and per capita NSDP as the two most influential variables, while UPI contributes much less. Figure 3 visualises that pattern, and Table 3 summarises the predictive, explainability and clustering results.

The clustering result gives the statistics a more practical reading. Rather than a neat national continuum, the data point to two broad development settings. One cluster combines higher digital access with stronger CE outcomes,

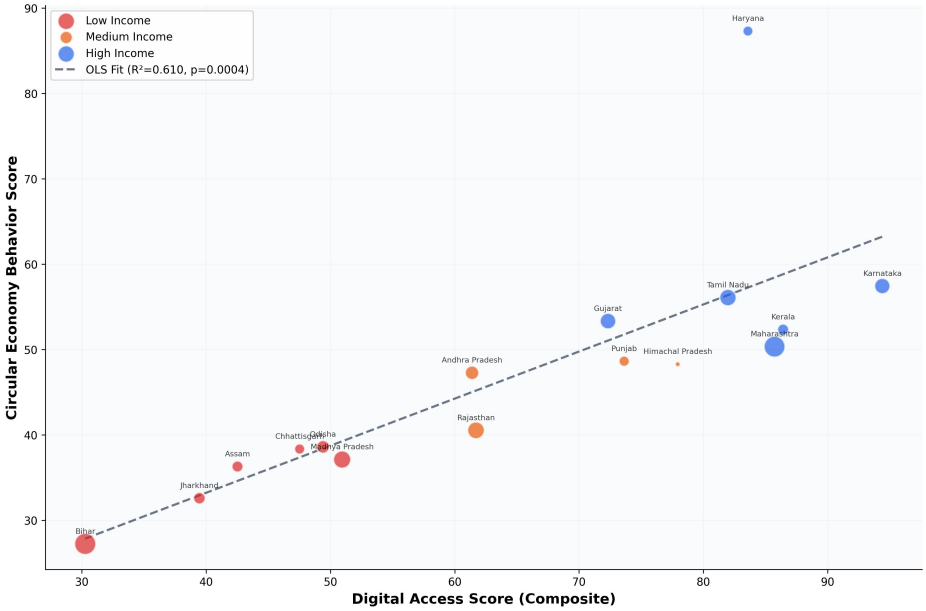


Figure 2: Digital Access Score and CE Behavior Score across the 16-state sample.

Source: Authors' analysis based on the compiled state-level dataset.

Table 2: Core Statistical Results

Test or Model	**Value**	**Interpretation**
OLS association (H1)	$R^2 = 0.610$; $\beta_1 = 0.551$; $p = 0.000352$	Strong positive relationship between DAS and CEBS
Pearson correlation	$r = 0.781$	Substantively large positive association
Spearman correlation	$\rho = 0.912$; $p < 0.001$	Rank-order relationship remains strong
HC3 robust inference	$p = 0.000011$	Result remains significant under heteroskedasticity-robust errors
Permutation test	$p = 0.0001$	Association is unlikely under random reassignment
Mediation via UPI (H2)	Sobel $z = -0.241$; $p = 0.810$	No evidence that UPI mediates the main relationship
Threshold analysis (H3)	Breakpoint $= 118.77\%$ tele-density	CE performance rises sharply up to the threshold, but evidence above it is based on very few states

Source: Authors' analysis.

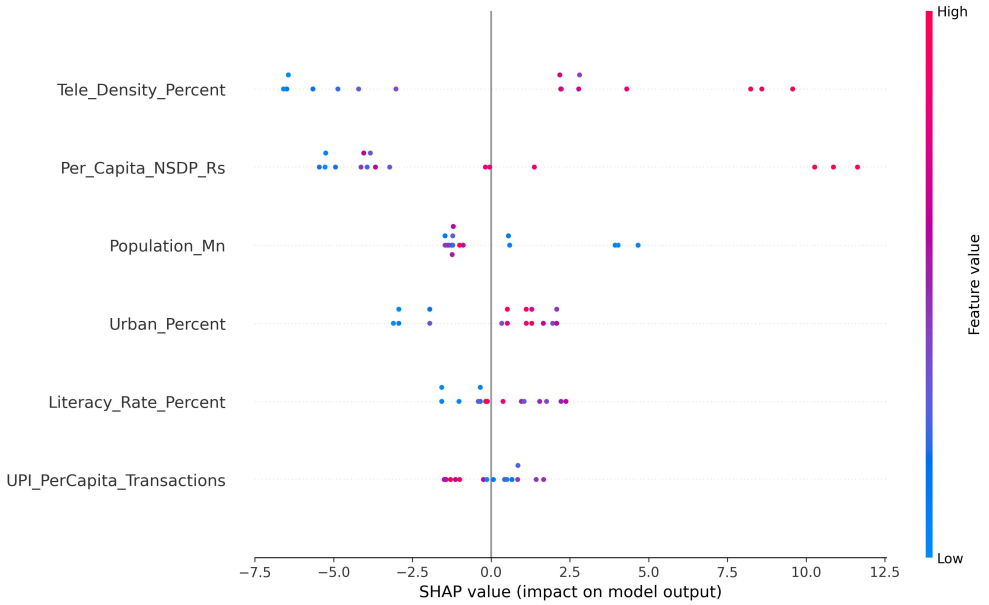


Figure 3: SHAP beeswarm plot for the nested XGBoost model.

Source: Authors' analysis.

Table 3: Predictive, Explainability and Clustering Summary

Analysis	**Key Result**	**Interpretation**
Best digital-only LOO-CV model	ElasticNet, $R^2 = 0.364$	Digital-access variables alone retain meaningful out-of-sample predictive value
Best full-model LOO-CV model	Lasso, $R^2 = 0.568$	Predictive performance remains close to the main OLS result
Best nested LOO-CV model	XGBoost, $R^2 = 0.511$	Main signal survives stricter leak-free evaluation
Top SHAP drivers	Tele-density; per capita NSDP	Infrastructure and economic capacity dominate the model explanation
K-Means typology	Two clusters; mean DCI 46.7 vs. 18.3	States separate into digital leaders and emerging states with a large circularity gap
Extended-sample check	$n = 21$; $r = 0.963$	Removing e-waste from CEBS strengthens rather than weakens the pattern

Source: Authors' analysis.

while the other remains weaker on both. This supports the view that digital infrastructure, administrative capacity and sustainability behavior are linked in practice. It also reinforces the point from the mediation test: tele-density matters more than digital payments alone.

4.4 Implications and limitations

The policy implication is not that connectivity automatically produces circularity. The more careful reading is that digital inclusion and CE strategy should not be planned as isolated domains. In lagging states, connectivity investment may also work as environmental capacity-building if it improves reporting, traceability, service delivery and administrative coordination. The study is still limited by its small cross-sectional sample, the use of 2011 Census controls, the inclusion of NSDP in both composite scores and the compression of within-state variation into state averages. The results therefore establish a consistent association, not a causal effect. Future work should extend the framework using district-level or panel data and broader circular-economy indicators.

5 Conclusion

This paper develops a Digital Circularity Index for 16 Indian states and finds a consistent association between digital access and circular-economy behavior. The result appears in regression, robustness checks, predictive validation and explainability analysis. Tele-density and economic capacity carry the strongest signal, while UPI adoption does not mediate the main relationship. The main lesson is therefore narrower, but more useful: digital inclusion should be understood not only as a communications or payments issue, but also as part of the institutional base on which better sustainability administration can be built.

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